Problem · Topic 3 · 4 points · 55 min

P16 — HESECA: the statements from a summary of ledger accounts, with prepaid insurance and a credit line

HELIPUERTOS Y SERVICIOS DEL CANTÁBRICO S.L. (HESECA) *operates one helicopter from Santander for charter flights, offshore inspections and maintenance of other operators' aircraft. A trainee of the accounting firm that keeps its books has prepared, instead of the financial statements, the following summary of the ledger accounts of 2022, from 1 January 2022 (01/01/2022) to 31 December 2022 (31/12/2022). The partner of the firm asks you to prepare the balance sheet and the Income Statement as of 31 December 2022 (31/12/2022).*

Opening balances at 1 January 2022 (01/01/2022): *Customers 3.000; Spare parts 1.000; Bank 2.000; Helicopter (cost) 24.000; Accumulated depreciation of the helicopter 4.500; Van (cost) 1.800; Accumulated depreciation of the van 300; Paid-in capital 18.000; Reserves 1.500; Suppliers 1.200; Unpaid taxes 800; Bank loan 5.500.*

Customers account: *opening balance 3.000; invoiced in 2022 for flights and maintenance 34.000; collected in 2022 31.000; on 1 September 2022 (01/09/2022) a balance of 400 m.u. from an operator declared in bankruptcy was written off as definitively unrecoverable.*

Spare parts and suppliers accounts: *on 1 March 2022 (01/03/2022) parts were purchased for 3.000 m.u., 50% paid in cash and the rest payable to 60 days; on 1 September 2022 (01/09/2022) parts were purchased for 3.000 m.u. more, 50% payable to 30 days and the other 50% deferred until 1 March 2023 (01/03/2023). The suppliers' balance of the previous year was paid in January. The physical count of 31 December 2022 (31/12/2022) values the stock (stock in stock) of parts at 1.400 m.u.*

Expense accounts (all paid by bank unless indicated): *fuel 4.800; salaries of pilots and mechanics 9.600; Social Security 3.000, of which the contribution of December (250) is paid in January 2023; hangar rent 200 m.u. / month; on 1 October 2022 (01/10/2022) the annual insurance premium of the helicopter, 1.200 m.u., covering from 1 October 2022 (01/10/2022) to 30 September 2023 (30/09/2023), was paid in full. The unpaid taxes of 2021 were paid in January.*

Fixed assets: *the helicopter is amortized linearly in 12 years with a residual value of 10% of its cost, and was bought on a date that the trainee could not find; the van was bought on 1 January 2021 (01/01/2021) and is amortized linearly in 6 years with residual value zero.*

Financing: *the bank loan was granted on 1 January 2020 (01/01/2020) for 7.500 m.u. at 6% annual interest; every 31 December the company returns 1.000 m.u. and pays the interest calculated on the debt at the beginning of the year. On 1 May 2022 (01/05/2022) a bank opened a credit line with a limit of 4.000 m.u.: HESECA drew 2.000 m.u. that day and 1.000 m.u. more on 1 September 2022 (01/09/2022); it pays 9% annual interest only on the amount drawn, every 31 July, 31 October and 31 January, for the three months then ended.*

Closing: *the results are declared at 23:59 hrs on 31 December 2022 (31/12/2022). The corporation tax is 30% and will be paid in July 2023 ("taxes to pay"). The partners decided on 31 December 2022 (31/12/2022) a dividend of 30% of the profit after taxes, to be paid on 15 March 2023 (15/03/2023); the rest goes to reserves.*

Required:

1. *Deduce the exact date on which the helicopter was purchased, and construct the Income Statement (Profit and Loss Account) of HESECA for the year 2022.*
2. *Calculate the bank balance at 31 December 2022 (31/12/2022) and construct the balance sheet at that date, justifying the classification of every item.*
3. *Determine the values of the ROA and ROE ratios in 2022, and indicate the effect on the balance sheet if the dividend had been paid at 24:00 on 31 December 2022 (31/12/2022).*
4. *The trainee had recorded the whole insurance premium of 1.200 m.u. as an expense of 2022. Indicate what changes this error would produce in the Income Statement, the distribution of the profit and the balance sheet, and explain which statement it distorts and why.*

Setup

Reading the ledger: a ledger account shows the opening balance, the increases and the decreases of the year; its closing balance is opening + increases − decreases. The balance sheet takes the closing balances of the asset, liability and equity accounts; the Income Statement takes the movements of the revenue and expense accounts.

Hypotheses:

1. The rent is paid every month by bank (12 × 200).
2. The insurance premium covers twelve months; only October–December 2022 are an expense of 2022, the rest is a **prepaid expense**, a current asset [Slide 14].
3. Credit-line interest: May–July is paid on 31/07, August–October on 31/10, November–December accrue and are paid on 31/01/2023.

Extension (not defined in the course decks). $\mathrm{ROA} = \text{Income from operations}/\mathrm{TA}$; $\mathrm{ROE} = \text{Net profit}/E$, on the balance at 31/12/2022.

Part (a) · 1.25 points

Deduce the purchase date of the helicopter and prepare the 2022 Income Statement

Step 1. Purchase date of the helicopter

$$D_{\text{helicopter}} = \frac{24.000 - 0{,}10 \times 24.000}{12} = \frac{21.600}{12} = 1.800 \text{ m.u./year}$$

$$\frac{4.500}{1.800} = 2{,}5 \text{ years before } 01/01/2022 \;\Rightarrow\; \text{bought on 1 July 2019 } (01/07/2019)$$

Check of the van: one year (2021) at $1.800/6 = 300$ gives the 300 of the opening balance (checked).

Step 2. Parts consumed

$$\text{Consumption} = 1.000 + 3.000 + 3.000 - 1.400 = 5.600 \text{ m.u.}$$

Step 3. Insurance (accrual)

$$\text{Expense } 2022 = 1.200 \times \frac{3}{12} = 300, \qquad \text{Prepaid at } 31/12/2022 = 1.200 - 300 = 900 \text{ m.u.}$$

Step 4. Interest

$$I_{\text{loan}} = 0{,}06 \times 5.500 = 330 \text{ m.u.}$$

$$I_{\text{credit}} = 2.000 \times 0{,}09 \times \frac{8}{12} + 1.000 \times 0{,}09 \times \frac{4}{12} = 120 + 30 = 150 \text{ m.u.}$$

Check of the loan: $7.500 - 2 \times 1.000 = 5.500$ at 01/01/2022 (checked).

Step 5. Income Statement 2022

Line	Computation	m.u.
Revenue from flights and maintenance	Customers account	34.000
– Spare parts consumed	Step 2	5.600
– Fuel		4.800
– Salaries		9.600
– Social Security		3.000
– Hangar rent	12×200	2.400

Line	Computation	m.u.
– Insurance (October–December)	Step 3	300
– Bad debt loss	Customers account	400
– Depreciation	$1.800 + 300$	2.100
= Income from operations	$34.000 - 28.200$	**5.800**
– Financial expenses	$330 + 150$	480
= Profit before taxes	$5.800 - 480$	**5.320**
– Corporation tax (30%)	$0{,}30 \times 5.320$	1.596
= Net profit	$5.320 - 1.596$	**3.724**

KEY RESULTS

- Helicopter: 1.800 m.u./year; bought on 1 July 2019 (01/07/2019).
- Parts consumed 5.600; insurance 300; depreciation 2.100 m.u.
- Income from operations 5.800; interest 330 + 150 = 480 m.u.
- Profit before taxes 5.320; tax 1.596; net profit 3.724 m.u.

Part (b) · 1.25 points

Compute the bank balance and prepare the balance sheet at 31 December 2022

Step 1. Closing the customers and suppliers accounts

$$\text{Customers}_{31/12} = 3.000 + 34.000 - 31.000 - 400 = 5.600 \text{ m.u.}$$

Supplier payment	Date	m.u.
Balance of 2021	January	1.200
50% of the March purchase	01/03/2022	1.500
50% of the March purchase, at 60 days	30/04/2022	1.500
50% of the September purchase, at 30 days	01/10/2022	1.500
Total paid		**5.700**

$$\text{Suppliers}_{31/12} = 1.200 + 6.000 - 5.700 = 1.500 \text{ m.u. (due } 01/03/2023)$$

Step 2. Credit-line interest paid

$$31/07:\ 2.000 \times 0{,}09 \times \frac{3}{12} = 45; \qquad 31/10:\ 2.000 \times 0{,}09 \times \frac{1}{12} + 3.000 \times 0{,}09 \times \frac{2}{12} = 15 + 45 = 60$$

$$\text{Paid} = 45 + 60 = 105, \qquad \text{Payable (November–December)} = 3.000 \times 0{,}09 \times \frac{2}{12} = 45 \text{ m.u.} \quad (105 + 45 = 150\ \checkmark)$$

Step 3. Bank account 2022

Inflows	**m.u.**	**Outflows**	**m.u.**
Collections from customers	31.000	Suppliers	5.700
Credit line drawn (2.000 + 1.000)	3.000	Fuel	4.800
		Salaries	9.600
		Social Security (3.000 − 250)	2.750
		Hangar rent	2.400
		Insurance premium	1.200
		Credit-line interest	105
		Loan interest	330
		Loan principal	1.000
		Unpaid taxes 2021	800
Total	**34.000**	**Total**	**28.685**

$$\text{Bank}_{31/12/2022} = 2.000 + 34.000 - 28.685 = 7.315 \text{ m.u.}$$

Step 4. Distribution

$$\text{Dividend} = 0{,}30 \times 3.724 = 1.117{,}20 \text{ (payable } 15/03/2023), \qquad \text{Reserves} = 1.500 + 3.724 - 1.117{,}20 = 4.106{,}80$$

Step 5. Balance sheet at 31/12/2022

Assets	m.u.	Liabilities and owners equity	m.u.
Helicopter (24.000 − 6.300)	17.700	Paid-in capital	18.000
Van (1.800 − 600)	1.200	Reserves	4.106,80
Non-current assets	**18.900**	**Owners equity**	**22.106,80**
Spare parts	1.400	Bank loan due 2024–2027	3.500
Customers	5.600	**Non-current liabilities**	**3.500**
Prepaid insurance	900	Bank loan, installment 31/12/2023	1.000
Bank	7.315	Credit line drawn	3.000
		Interest payable (credit line)	45
		Suppliers	1.500
		Social Security payable	250
		Taxes to pay	1.596
		Dividends payable	1.117,20
Current assets	**15.215**	**Current liabilities**	**8.508,20**
Total assets	**34.115**	**Total liabilities and equity**	**34.115**

The loan: $5.500 - 1.000 = 4.500$ m.u., of which 1.000 is due on 31/12/2023. The credit line is a short-term bank credit [Slide 31]; the prepaid insurance is a current asset because it will be consumed within the year [Slide 14].

KEY RESULTS

- Customers 5.600; suppliers 1.500; payments to suppliers 5.700 m.u.
- Credit-line interest paid 105, payable 45 m.u.
- Bank balance 7.315 m.u.
- Owners equity 22.106,80; current liabilities 8.508,20; total 34.115 m.u.

Part (c) · 0.75 points

Compute ROA and ROE, and the effect of paying the dividend at 24:00 on 31 December 2022

$$\text{ROA} = \frac{5.800}{34.115} = 17{,}00\%, \qquad \text{ROE} = \frac{3.724}{22.106{,}80} = 16{,}85\%$$

Item	Paid 15/03/2023	Paid at 24:00	Change
Bank	7.315	6.197,80	-1.117,20
Dividends payable	1.117,20	0	-1.117,20
Owners equity	22.106,80	22.106,80	0
Total assets = total liabilities and equity	34.115	32.997,80	-1.117,20

KEY RESULTS

- ROA = 17,00%; ROE = 16,85% (extension).
- Paid at 24:00: bank 6.197,80; dividends payable 0; totals 32.997,80 m.u.

Part (d) · 0.75 points

The trainee's error: the whole insurance premium as an expense of 2022

Step 1. Effect on the results

$$\text{Profit before taxes} = 5.320 - 900 = 4.420, \qquad \text{Tax} = 0{,}30 \times 4.420 = 1.326, \qquad \text{Net profit} = 4.420 - 1.326 = 3.094$$

$$\text{Dividend} = 0{,}30 \times 3.094 = 928{,}20, \qquad \text{Reserves} = 1.500 + 3.094 - 928{,}20 = 3.665{,}80$$

Step 2. Effect on the balance sheet

Item	Correct	With the error	Change
Prepaid insurance	900	0	-900
Total assets	**34.115**	**33.215**	**-900**

Item	Correct	With the error	Change
Reserves	4.106,80	3.665,80	-441
Taxes to pay	1.596	1.326	-270
Dividends payable	1.117,20	928,20	-189
Total liabilities and equity	**34.115**	**33.215**	**-900**

Check: $441 + 270 + 189 = 900$ (checked).

Step 3. Which statement the error distorts

The bank balance is the same: the 1.200 m.u. were paid either way. The error is an **accrual** error: it charges to 2022 nine months of insurance that belong to 2023 [Slide 8]. The **Income Statement** of 2022 understates the profit by 630 m.u. (and that of 2023 will overstate it), and the **balance sheet** omits a resource the company still owns, the cover of January–September 2023. It also understates the tax and the dividend, so the partners receive 189 m.u. less than they should.

KEY RESULTS

- Insurance expense 1.200 (+900); profit before taxes 4.420; tax 1.326 (-270); net profit 3.094 (-630).
- Dividend 928,20 (-189); reserves 3.665,80 (-441).
- Prepaid insurance disappears (-900); bank unchanged.

Interpretation and checks

- **Dimensional check:** m.u. × rate × months/12 gives m.u.; ratios dimensionless.
- **Ledger logic:** every closing balance follows from opening + increases − decreases; the customers account alone gives revenue (increase), collections and the bad debt (decreases) and the receivable (closing balance).
- **Profit vs cash:** net profit 3.724 m.u., while the bank rose by 5.315 m.u.: $3.724 + 2.100$ (depreciation) $+3.000$ (credit line) -1.000 (principal) -900 (prepaid insurance) -2.600 (rise in customers) -400 (rise in parts) $+300 + 250 + 796 + 45$ (more owed to suppliers, Social Security, tax and interest) $= 5.315$.
- **Typical mistakes:** expensing the whole insurance, charging the credit-line interest on the 4.000 m.u. limit instead of the amount drawn, treating the dividend decided as already paid, and forgetting the bad debt in the customers account.